

AeroNetB ASCM Implementation Logbook

5CM506 Data Driven Systems - D-II

Field	Value
Student Name	Spyros Kokkoris
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System	AeroNetB Aerospace Supply Chain Management Dashboard
Deployment Mode	Cloud deployment through Render Blueprint, with local fallback for testing

1. Implementation Details

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Role-based dashboard with charts, search, forms, and CSV export.
Backend	Node.js HTTP API	Authentication, RBAC, validation, cross-database responses, and audit logging.
Relational DB	PostgreSQL on Render	Suppliers, parts, orders, shipments, employees, roles, equipment, and audit logs.
Document DB	MongoDB Atlas	QC reports, certifications, IoT sensor logs, and compliance flags.
Deployment	Render Blueprint	Web service plus PostgreSQL database from render.yaml.

1.1 Access Instructions

Item	Value
GitHub repository	https://github.com/Spyko2001/aeronetb-ascm-dashboard
Live Render app URL	https://aeronetb-ascm-dashboard.onrender.com
Live health endpoint	https://aeronetb-ascm-dashboard.onrender.com/api/health
Verified database mode	PostgreSQL + MongoDB, database_warning: null
PostgreSQL connection	Managed by Render through DATABASE_URL; private value is not included.
MongoDB connection	Managed by Render through MONGODB_URI; private value is not included.

1.2 Demo Accounts

Role	Auth ID	Password
Procurement Officer	priya.procurement	demo123
Quality Inspector	lena.inspector	demo123
Supply Chain Manager	marcus.manager	demo123
Equipment Engineer	irene.engineer	demo123
Auditor / Regulator	omar.auditor	demo123

2. Refined Logical Models

The first-semester design was refined into a deployable hybrid model. PostgreSQL stores

stable structured domains, while MongoDB stores flexible document domains. The backend validates cross-database references and combines records for the dashboard.

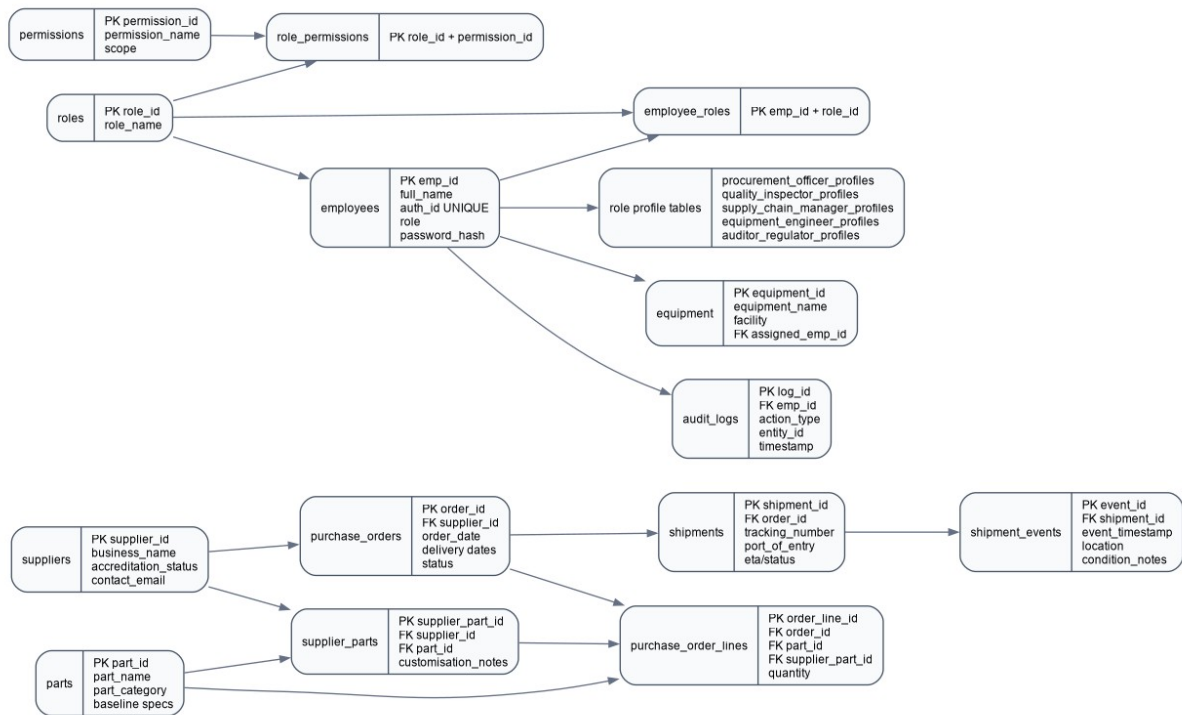


Figure 1 - Refined PostgreSQL relational schema

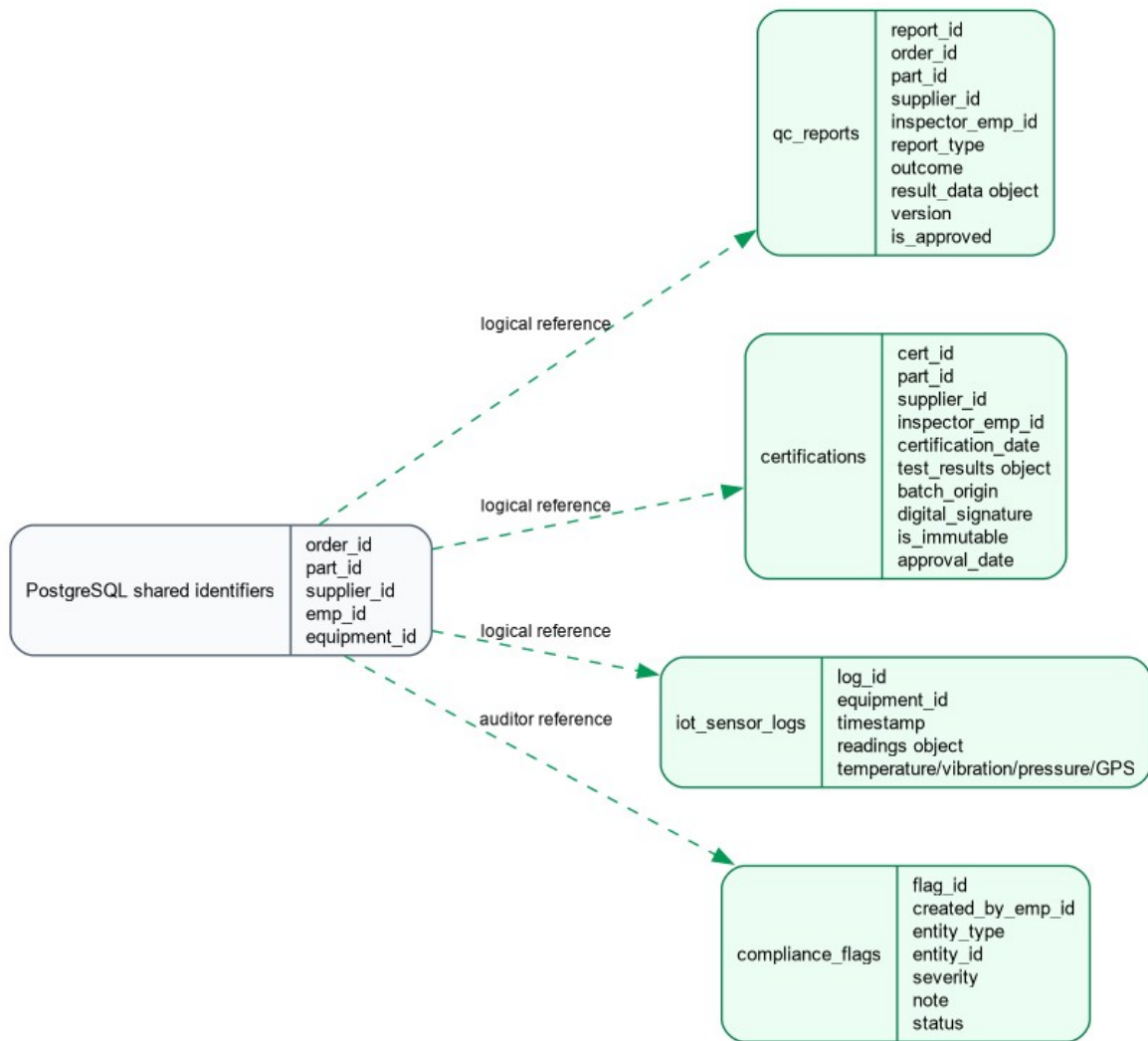


Figure 2 - Refined MongoDB document model

2.1 Refinements From Task 1

Refinement	Implementation
Explicit RBAC model	Added roles, permissions, role_permissions, and employee_roles tables, with backend permission checks.
Versioned QC reports	New QC submissions create a new MongoDB document version instead of overwriting previous results.
Immutable certifications	Approved certifications have <code>isImmutable=true</code> and later updates return HTTP 409.
Audit coverage	Login, view, create, update, approve, export, and flag actions create audit entries with <code>emp_id</code> .
Deployment readiness	Render Blueprint, environment variables, automatic schema creation, automatic seeding, and local fallback.

3. Database Implementation

Evidence File	Description
sql_scripts/ddl.sql	PostgreSQL table definitions, keys, constraints, and indexes.

sql_scripts/dml.sql	Dummy relational data for suppliers, parts, orders, shipments, employees, roles, profiles, equipment, and audit logs.
mongo_scripts/collections.js	MongoDB collection creation, JSON Schema validators, and indexes.
backend/src/seed-data.js	Runtime seed records used by the Node application for PostgreSQL and MongoDB bootstrap.

4. API Design and Development

Method	Endpoint	Description	Data Source
POST	/api/auth/login	Authenticates a seeded user and returns a signed token.	PostgreSQL employees
GET	/api/dashboard	Returns KPIs, alerts, supplier performance, QC summary, shipments, and IoT status.	PostgreSQL + MongoDB
GET/POST	/api/suppliers	Lists or creates suppliers.	PostgreSQL
GET/POST/PATCH	/api/orders	Lists, creates, or updates purchase orders.	PostgreSQL
GET/POST	/api/qc-reports	Lists reports or creates a new versioned QC report.	MongoDB
GET/POST/PATCH	/api/certifications	Lists, creates, or updates pending certifications.	MongoDB
GET/POST	/api/iot	Lists or submits telemetry readings.	MongoDB
GET	/api/audit-logs	Lists recent audit trail events.	PostgreSQL

5. Dashboard Evidence

Overview

Procurement
Shipments
Quality
Equipment
Compliance
Audit

Operational Overview

Live dashboard covering supplier performance, shipments, quality, certifications, and IoT alerts.

SUPPLIERS

4

OPEN ORDERS

3

DELAYED

1

FAILED QC

2

PENDING CERTS

1

IOT WARNINGS

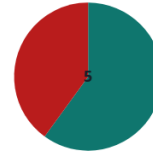
2

Supplier performance



QC outcomes

PASS: 3
FAIL: 2



Alerts

- High** **Delayed order**
Order 5002 from Aegean Precision Alloys is past desired delivery.
- High** **QC failure**
QC-2026-00004 failed for A320 fuselage panel from Aegean Precision Alloys.
- High** **QC failure**
QC-2026-00003 failed for A320 fuselage panel from Aegean Precision Alloys.
- Medium** **IoT threshold**
Autoclave AC-17 is warning based on latest sensor reading.
- Medium** **IoT threshold**
Container Tracker CT-44 is warning based on latest sensor reading.
- Medium** **Certification pending**
CERT-2026-00002 is not immutable yet

Figure 3 - Supply Chain Manager overview dashboard with KPIs and charts



Overview

Procurement

Shipments

Quality

Compliance

Procurement

Create suppliers and orders, then track the order lifecycle.

Add supplier

Business name

Accreditation

Contact email

Contact name

Phone

Address

Add supplier

Create purchase order

Supplier

Part

Quantity

Desired delivery

Status

Create order

Suppliers

ID	BUSINESS	ACCREDITATION	CONTACT
2	Aegean Precision Alloys	ISO 9001	Nikos Markou orders@aegean-alloys.example
1	Helios Aero Composites	AS9100, ISO 9001	Klara Weber supply@helios-aero.example
3	NorthStar Avionics Materials	AS9100	Maya Deschamps quality@northstar-materials.example
4	Pacific AeroForge	Pending renewal	Lee Wei Chen sales@pacific-aeroforge.example

Orders

Figure 4 - Procurement view for supplier and order workflow



Overview

Procurement

Shipments

Quality

Compliance

Quality Control

Create versioned QC reports and lock certifications after approval.

Create QC report

Order	Supplier	Part
Order 5005	Aegean Precision Alloys	A320 fuselage panel
Report type	Outcome	
visual	PASS	
Notes		
inspection result details	Create QC report	

Create certification

Supplier	Part	Batch origin
Aegean Precision Alloys	A320 fuselage panel	
Test summary	Digital signature	
PENDING	SHA256:pending	Create certification

QC reports

REPORT	SUPPLIER	PART	TYPE	OUTCOME	APPROVED	ACTION
QC-2026-00004 v2	Aegean Precision Alloys	A320 fuselage panel	ndt	FAIL	pending	Approve
QC-2026-00005 v1	Pacific AeroForge	Composite landing gear door	environmental	PASS	pending	Approve
QC-2026-00003 v1	Aegean Precision Alloys	A320 fuselage panel	ndt	FAIL	pending	Approve
QC-2026-00002 v1	Aegean Precision Alloys	Titanium fastener kit	visual	PASS	PASS	-
QC-2026-00001	Helios Aero Composites	A320 fuselage panel	dimensional	PASS	PASS	-

Figure 5 - Quality view for QC report and certification workflow



Overview

Shipments

Equipment

Equipment & IoT

Monitor sensor thresholds and submit new telemetry readings.

Add IoT reading

Equipment

Autoclave AC-17

Temperature C

72

Vibration mm/s

1.4

Pressure bar

3.8

Submit reading

Sensor trend



Machine status

Warning

Autoclave AC-17

Derby assembly plant

OK

CNC Cell M5

Derby assembly plant

Warning

Container Tracker CT-44

Shipment 7002

OK

CMM Inspection Bench Q2

Quality lab

IoT logs

LOG	EQUIPMENT	TIMESTAMP	TEMPERATURE	VIBRATION	STATUS
IOT-2026-00005	Container Tracker CT-44	May 21, 2026, 01:20 PM	7.2	2.9	Warning
IOT-2026-00004	CNC Cell M5	May 21, 2026, 12:30 PM	69.4	2.7	OK
IOT-2026-00006	CMM Inspection Bench Q2	May 21, 2026, 12:00 PM	22	0.3	OK
IOT-2026-00002	Autoclave AC-17	May 21, 2026, 11:00 AM	78.2	1.4	Warning
IOT-2026-00003	CNC Cell M5	May 21, 2026, 10:30 AM	66.1	2.2	OK

Figure 6 - Equipment engineer IoT monitoring view



Overview

Procurement

Shipments

Quality

Equipment

Compliance

Audit

Compliance

Read-only certification review with auditor flagging for non-compliance cases.

Export certs

Raise compliance flag

Entity type

Certification

Entity ID

CERT-2026-00003

Severity

Medium

Note

Raise flag

Certification records

CERTIFICATE	SUPPLIER	PART	BATCH	APPROVED	APPROVAL DATE	ACTION
CERT-2026-00003	Pacific AeroForge	Composite landing gear door	Batch-PAF-2026-21	pending	-	-
CERT-2026-00001	Helios Aero Composites	A320 fuselage panel	Batch-HAC-2026-19	completed	May 14, 2026, 01:00 PM	-
CERT-2026-00002	Aegean Precision Alloys	Titanium fastener kit	Batch-APA-2026-44	completed	May 14, 2026, 04:30 PM	-

Compliance flags

FLAG	ENTITY	SEVERITY	NOTE	STATUS
1	Certification CERT-2026-00003	Medium	Seal integrity is still pending before final approval.	open

Figure 7 - Auditor compliance and certification review view

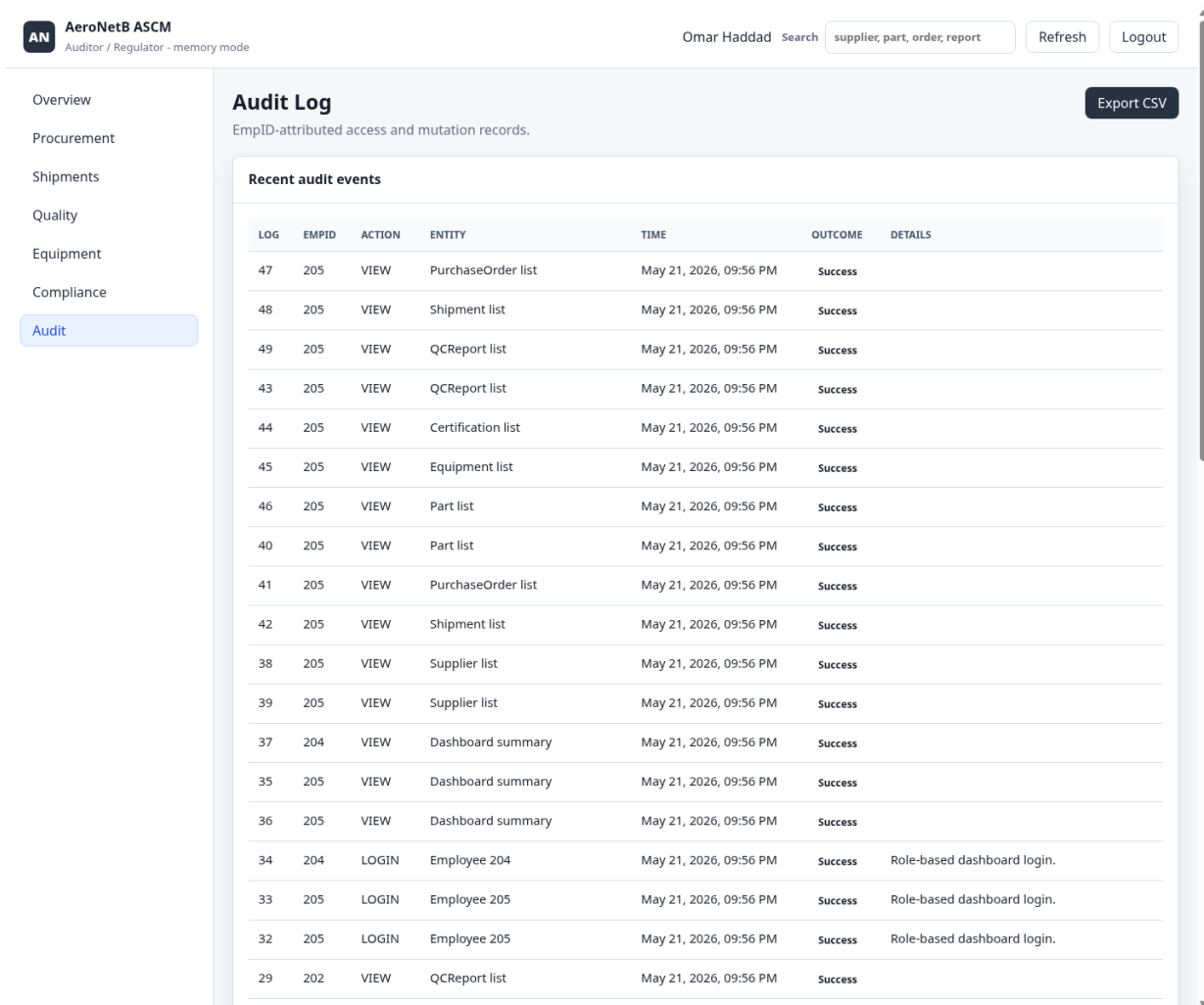


Figure 8 - Audit log with emp_id-attributed actions

6. Security and Compliance

Control	Implementation
Authentication	Users log in with auth ID and password. Stored passwords are PBKDF2 hashes with salts.
Session protection	The API issues signed HMAC session tokens with expiry.
RBAC	Every protected endpoint checks permissions in the backend before reading or mutating data.
Auditor restriction	The auditor can read compliance data and raise flags but cannot mutate suppliers, orders, QC reports, or certifications.
Certification immutability	Approved certifications have is_immutable=true. Later update attempts return HTTP 409.
Audit trail	Important actions write records with emp_id, action type, entity, timestamp, outcome, and details.

7. Testing and Validation

Test	Expected Result	Status
npm run check	JavaScript syntax checks pass.	Passed
npm test	Integration tests pass against a local seeded service.	Passed
Auditor creates supplier	Rejected with HTTP 403.	Passed

QC report retest	New report receives next version number.	Passed
Approved certification update	Rejected with HTTP 409.	Passed
CSV export	Manager can export audit logs.	Passed

8. Deployment Explanation

Render was chosen because it deploys the Node.js service directly from GitHub and creates a PostgreSQL database from the same Blueprint file. MongoDB Atlas stores flexible QC, certification, IoT, and compliance documents. The Render service uses environment variables for private connection strings.

9. Demo Flow

- Open the Render URL and show /api/health.
- Log in as Supply Chain Manager and show KPIs, alerts, charts, search, and CSV export.
- Log in as Procurement Officer and create a supplier or order.
- Log in as Quality Inspector and create/approve QC and certification records.
- Show certification immutability and QC versioning.
- Log in as Equipment Engineer and add an IoT reading.
- Log in as Auditor and show read-only compliance plus compliance flag creation.
- Show audit logs proving actions are attributed to emp_id.